

# LECTURE 2 – DYNAMICAL SYSTEMS

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INTRODUCTION TO AUTONOMOUS SYSTEMS · MODULE II

**Topics:** State–space models · Mechanics · Partial Observability

# AGENDA

| S | TOPIC                        | KEY IDEAS                                                                                 |
|---|------------------------------|-------------------------------------------------------------------------------------------|
| 1 | <b>The Tuple</b>             | $(\mathcal{X}, \mathcal{O}, \mathcal{A}, f, h, \pi)$ — components and 5 canonical systems |
| 2 | <b>Mechanics</b>             | Lagrangian derivation · distributed mass · manipulator equation                           |
| 3 | <b>Partial Observability</b> | Jacobian rank · collinear beacons · geometric diversity                                   |

## PREREQUISITES

- Linear algebra: eigenvalues, matrix exponential, positive-definite matrices
- ODEs: first-order systems, existence-uniqueness, separation of variables
- Classical mechanics: Newton's laws, kinetic/potential energy, torque
- Multivariable calculus: Jacobians, chain rule, gradient

| TOPIC                          | REFERENCE                                                                                                                                                    |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Linear algebra / matrix theory | Гантмахер — <i>Теорія матриць</i> (укр. вид.); Horn & Johnson — <i>Matrix Analysis</i> (2nd ed., 2013)                                                       |
| ODEs                           | Арнольд — <i>Звичайні диференціальні рівняння</i> (укр. вид.); Coddington & Levinson — <i>Theory of ODEs</i> (1955)                                          |
| Classical mechanics            | Ландау & Ліфшиць — <i>Механіка</i> (т. 1, укр. вид.); Goldstein, Poole, Safko — <i>Classical Mechanics</i> (3rd ed., 2002)                                   |
| Multivariable calculus         | Фіхтенгольц — <i>Курс диференціального та інтегрального числення</i> (т. 2–3, укр. вид.); Rudin — <i>Principles of Mathematical Analysis</i> (3rd ed., 1976) |

Українські підручники та матеріали: [nebayduzhi-math.azurewebsites.net](http://nebayduzhi-math.azurewebsites.net)

# LITERATURE

## Standard references

|                                                                                        |                                                            |
|----------------------------------------------------------------------------------------|------------------------------------------------------------|
| Strogatz — <i>Nonlinear Dynamics and Chaos</i> (2nd ed., 2015)                         | Intuitive intro; §2–6 cover everything in this lecture     |
| Khalil — <i>Nonlinear Systems</i> (3rd ed., 2002)                                      | Rigorous Lyapunov theory, Lipschitz, Picard–Lindelöf       |
| Murray, Li, Sastry — <i>A Mathematical Introduction to Robotic Manipulation</i> (1994) | Lagrangian mechanics, manipulator equations; free online   |
| Spong, Hutchinson, Vidyasagar — <i>Robot Modeling and Control</i> (2005)               | Mechanics §3–4, practical reference for §2 of this lecture |

## Advanced references

|                                                                                 |                                                   |
|---------------------------------------------------------------------------------|---------------------------------------------------|
| Marsden & Ratiu — <i>Introduction to Mechanics and Symmetry</i> (2nd ed., 1999) | Geometric / Hamiltonian mechanics on manifolds    |
| Bullo & Lewis — <i>Geometric Control of Mechanical Systems</i> (2004)           | Control theory on Lie groups; free online         |
| Bloch — <i>Nonholonomic Mechanics and Control</i> (2nd ed., 2015)               | Constraints, distribution theory, rolling contact |

# S1 · THE TUPLE

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*PRECISE DEFINITIONS OF EVERY COMPONENT*



# THE AUTONOMOUS SYSTEM TUPLE

**Definition.** A controlled dynamical system is specified by the six-tuple  $(\mathcal{X}, \mathcal{O}, \mathcal{A}, f, h, \pi)$ :

| SYMBOL        | NAME              | MEANING                                                                              |
|---------------|-------------------|--------------------------------------------------------------------------------------|
| $\mathcal{X}$ | State space       | Minimal info to predict future behaviour                                             |
| $\mathcal{O}$ | Observation space | What sensors return — a lossy projection of $\mathcal{X}$                            |
| $\mathcal{A}$ | Action space      | What the agent can apply                                                             |
| $f$           | Dynamics          | How the robot evolves: $\dot{x} = f(x, a)$ — comes from physics, <b>not designed</b> |
| $h$           | Observation model | How sensors project state: $o = h(x)$ — set by sensor physics                        |
| $\pi$         | Policy            | What the agent decides: $a = \pi(o)$ — the object we <b>design</b>                   |

Every lecture from now on **refines exactly one component** of this tuple.

## STATE SPACE

The **state**  $x_t \in \mathcal{X}$ : minimal information to predict all future behaviour given future actions.

$\mathcal{X}$  is typically:

- $\mathbb{R}^n$  — positions, velocities, temperatures
- A **Lie group**:  $SO(2)$ ,  $SO(3)$ ,  $SE(3)$  — systems with rotational degrees of freedom
- A finite set  $\{1, \dots, N\}$  — discrete/symbolic systems

The state lives in a **manifold**, not necessarily  $\mathbb{R}^n$ .

On  $SO(3)$ : you cannot subtract two rotation matrices to get a meaningful error vector.

**Rule of thumb (mechanical systems):** state = all generalised coordinates **and** all time derivatives.

## OBSERVATION SPACE AND OBSERVATION MODEL

The observation:  $o_t = h(x_t)$  — almost always **lossy** ( $h$  not injective).

| SENSOR           | $\mathcal{O}$                            | WHAT THE MODEL DISCARDS                               |
|------------------|------------------------------------------|-------------------------------------------------------|
| GPS              | $\mathbb{R}^3$ position                  | velocity, attitude, internal dynamics                 |
| Monocular camera | $\mathbb{R}^{H \times W \times 3}$ image | depth, occluded regions                               |
| LiDAR            | point cloud $\mathbb{R}^{N \times 3}$    | colour, texture                                       |
| IMU              | $(\dot{v}, \omega) \in \mathbb{R}^6$     | position, absolute orientation                        |
| Rotary encoder   | joint angle $q \in S^1$                  | joint velocity $\dot{q}$ — must be finite-differenced |

There is a gap between what the system *is* ( $x$ ) and what the agent *sees* ( $o$ ).

## ACTION SPACE

The **action**  $a_t \in \mathcal{A}$  is what the agent can apply to the system.

- **Continuous:**  $a \in \mathbb{R}^m$  — motor torques, voltages, force vectors
- **Discrete:** {left, right, forward, stop} — grid navigation
- **Mixed:** continuous actions subject to discrete mode switches (e.g. gear selection)

Constraints on  $\mathcal{A}$  (actuator limits, safety margins) are central to optimal control — Module IV.

## DYNAMICS: FOUR VARIANTS

| TYPE                              | EQUATION                                                                               |
|-----------------------------------|----------------------------------------------------------------------------------------|
| Deterministic, continuous time    | $\dot{x} = f(x, a), \quad f : \mathcal{X} \times \mathcal{A} \rightarrow T\mathcal{X}$ |
| Deterministic, discrete time      | $x_{t+1} = f(x_t, a_t)$                                                                |
| Stochastic, discrete time         | $x_{t+1} \sim p(\cdot \mid x_t, a_t)$                                                  |
| Stochastic, continuous time (SDE) | $dx = f(x, a) dt + \sigma(x, a) dW$                                                    |

For  $\mathcal{X} = \mathbb{R}^n$ :  $f$  maps to  $\mathbb{R}^n$ .

For  $\mathcal{X} = SO(3)$ :  $f(x, a)$  must lie in the **tangent space**  $T_R SO(3)$  at the current rotation  $R$ .  
 $T_R SO(3) \neq SO(3)$ .

$f$  comes from Lagrangian mechanics. It is **not designed**.

- Classical control → **Deterministic, continuous time**
- Reinforcement learning → **Stochastic, discrete time**
- Kalman filter → **Stochastic, continuous time, linear**

# POLICY

The **policy** maps what the agent observes to what it does:

| POLICY TYPE      | FORM                                 | NOTES                                                            |
|------------------|--------------------------------------|------------------------------------------------------------------|
| State-feedback   | $\pi(x_t)$                           | Requires full state — not directly realizable without estimation |
| Output-feedback  | $\pi(o_t)$                           | What the agent actually implements                               |
| History-feedback | $\pi(o_{1:t})$                       | Compensates for partial observability; belief-based              |
| Recurrent / LSTM | Compresses $o_{1:t}$ to hidden state | Practical approximation to history-feedback                      |

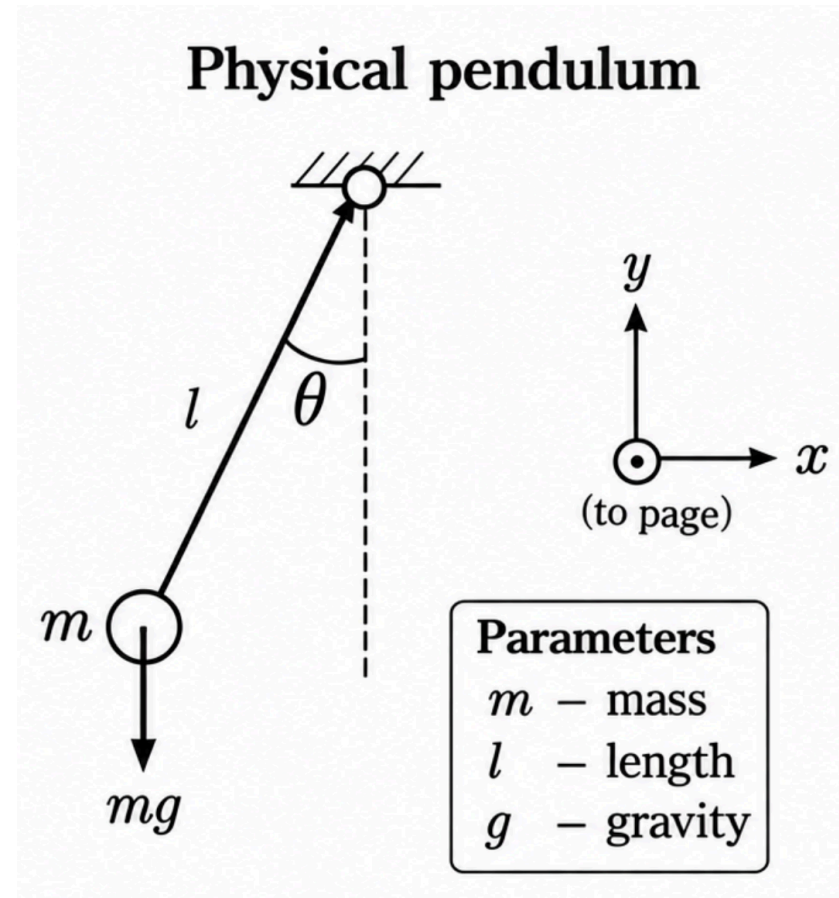
The distinction between  $\pi(x_t)$  and  $\pi(o_t)$  is not cosmetic.

Optimal state-feedback controllers exist by HJB (Module IV).

Extending to output-feedback requires estimation — Module III.

## PENDULUM: DEFINITION

A point mass  $m$  on a massless rod of length  $l$ , pinned at a fixed pivot.



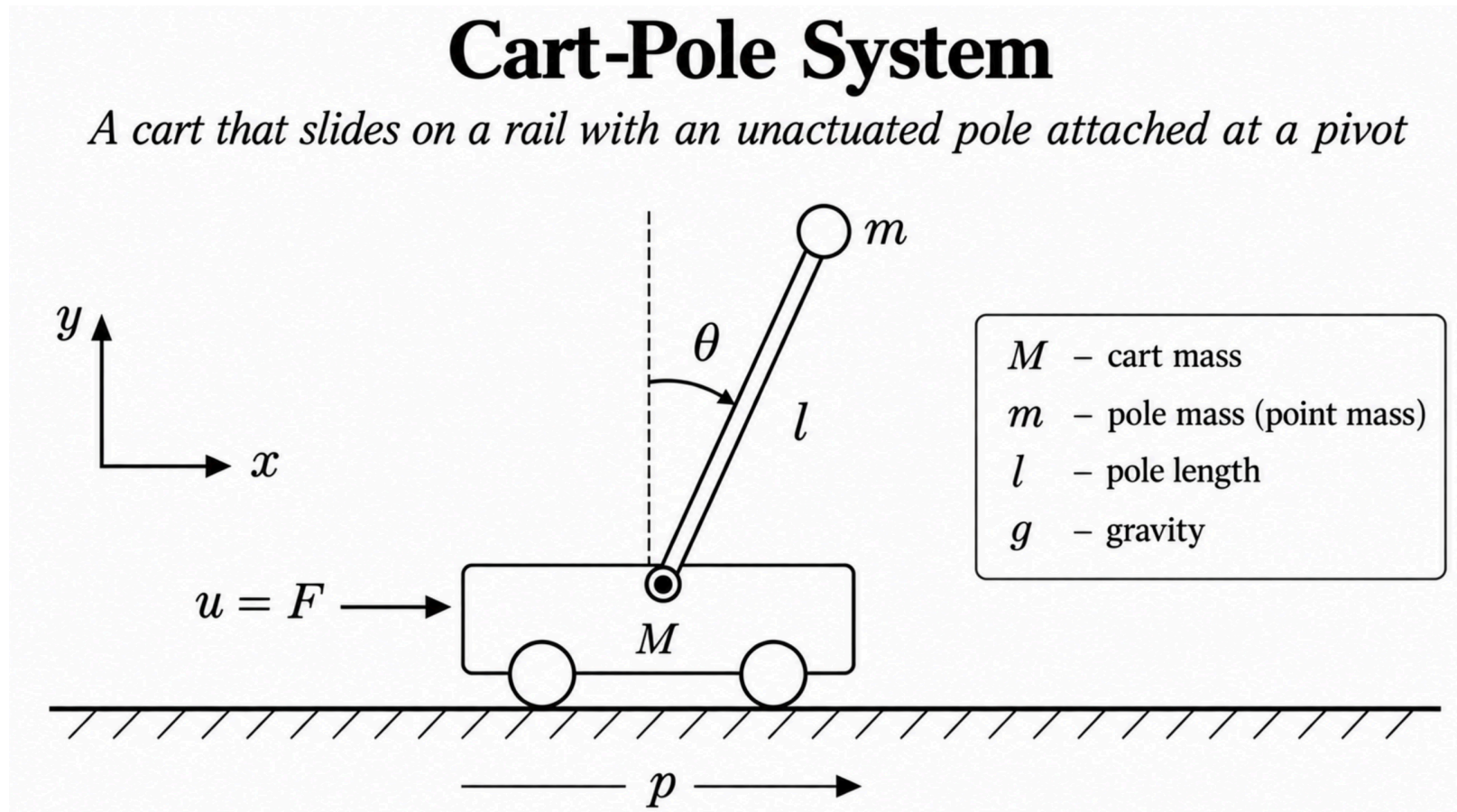
## PENDULUM: TUPLE

| ELEMENT       | SPECIFICATION                                                                            |
|---------------|------------------------------------------------------------------------------------------|
| $\mathcal{X}$ | $(\theta, \dot{\theta}) \in S^1 \times \mathbb{R}$ , dim 2 — angle and angular velocity  |
| $\mathcal{O}$ | Rotary encoder: $o = \theta \in S^1$ , dim 1 — $\dot{\theta}$ must be finite-differenced |
| $\mathcal{A}$ | Torque at pivot: $a = \tau \in \mathbb{R}$ ; uncontrolled: $\tau = 0$                    |
| $f$           | $f(x, a) = (\dot{\theta}, -(g/l) \sin \theta + \tau/(ml^2))$                             |
| $h$           | $h(\theta, \dot{\theta}) = \theta$ — velocity is a hidden component of the state         |
| $\pi$         | To design: energy shaping (swing-up), LQR (balance at inverted equilibrium)              |

**Key:**  $h$  discards  $\dot{\theta}$ . The agent sees angle only.

## CART-POLE: DEFINITION

**Cart-pole** — cart on rail with unactuated pole

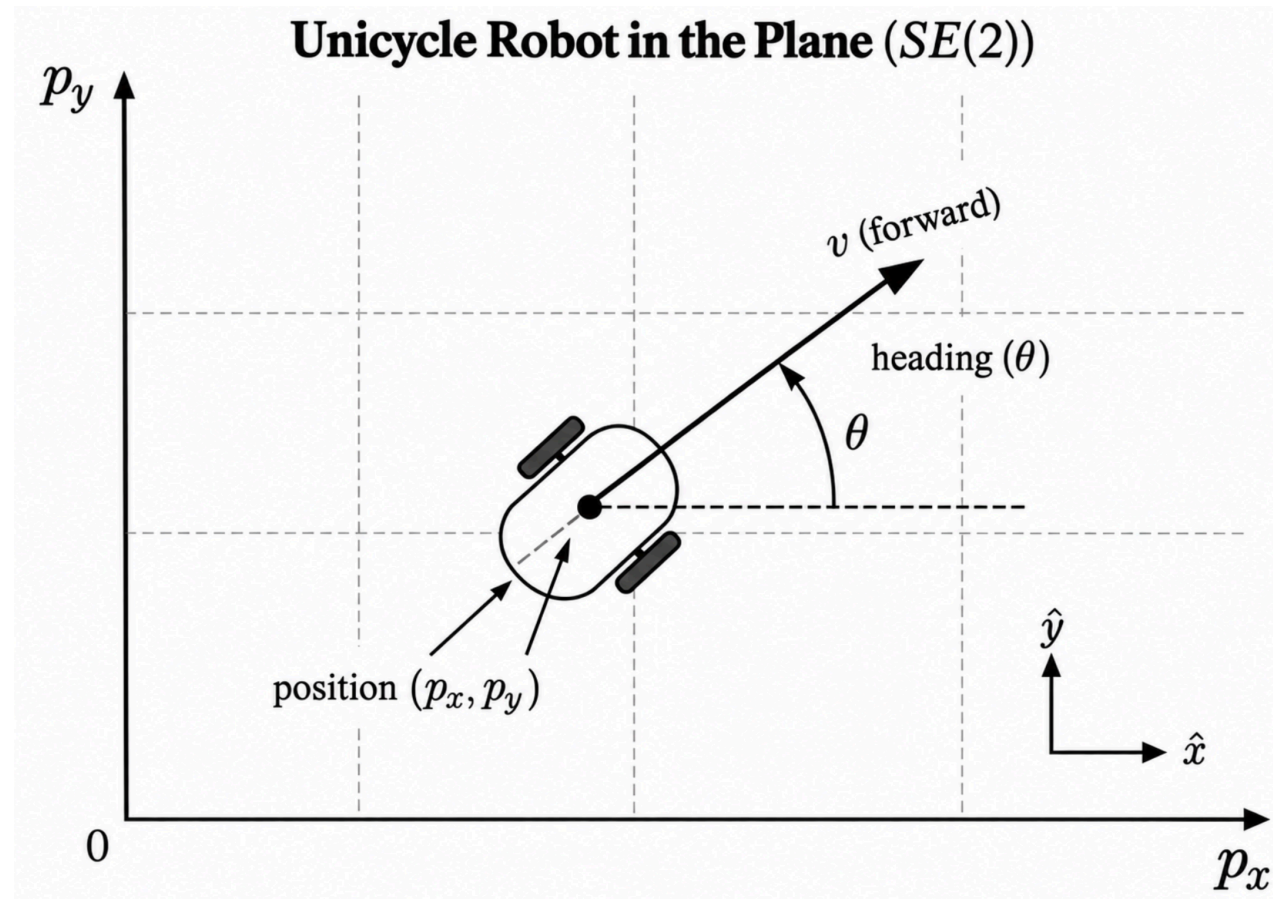


# CART-POLE: TUPLE

| ELEMENT       | SPECIFICATION                                                                              |
|---------------|--------------------------------------------------------------------------------------------|
| $\mathcal{X}$ | $(p, \theta, \dot{p}, \dot{\theta}) \in \mathbb{R} \times S^1 \times \mathbb{R}^2, \dim 4$ |
| $\mathcal{O}$ | Rail encoder $p$ + pole encoder $\theta$ : $\dim 2$ — <b>both velocities hidden</b>        |
| $\mathcal{A}$ | Horizontal force on cart: $F \in \mathbb{R}$ — pole <b>unactuated</b>                      |
| $f$           | 2 coupled nonlinear ODEs: one for cart $p$ , one for pole $\theta$                         |
| $h$           | $h(p, \theta, \dot{p}, \dot{\theta}) = (p, \theta)$ — discards both velocities             |
| $\pi$         | To design: swing-up + balance; benchmark for underactuated nonlinear control               |

## UNICYCLE: DEFINITION

**Unicycle robot** — wheeled robot with non-holonomic (no-slip) constraint

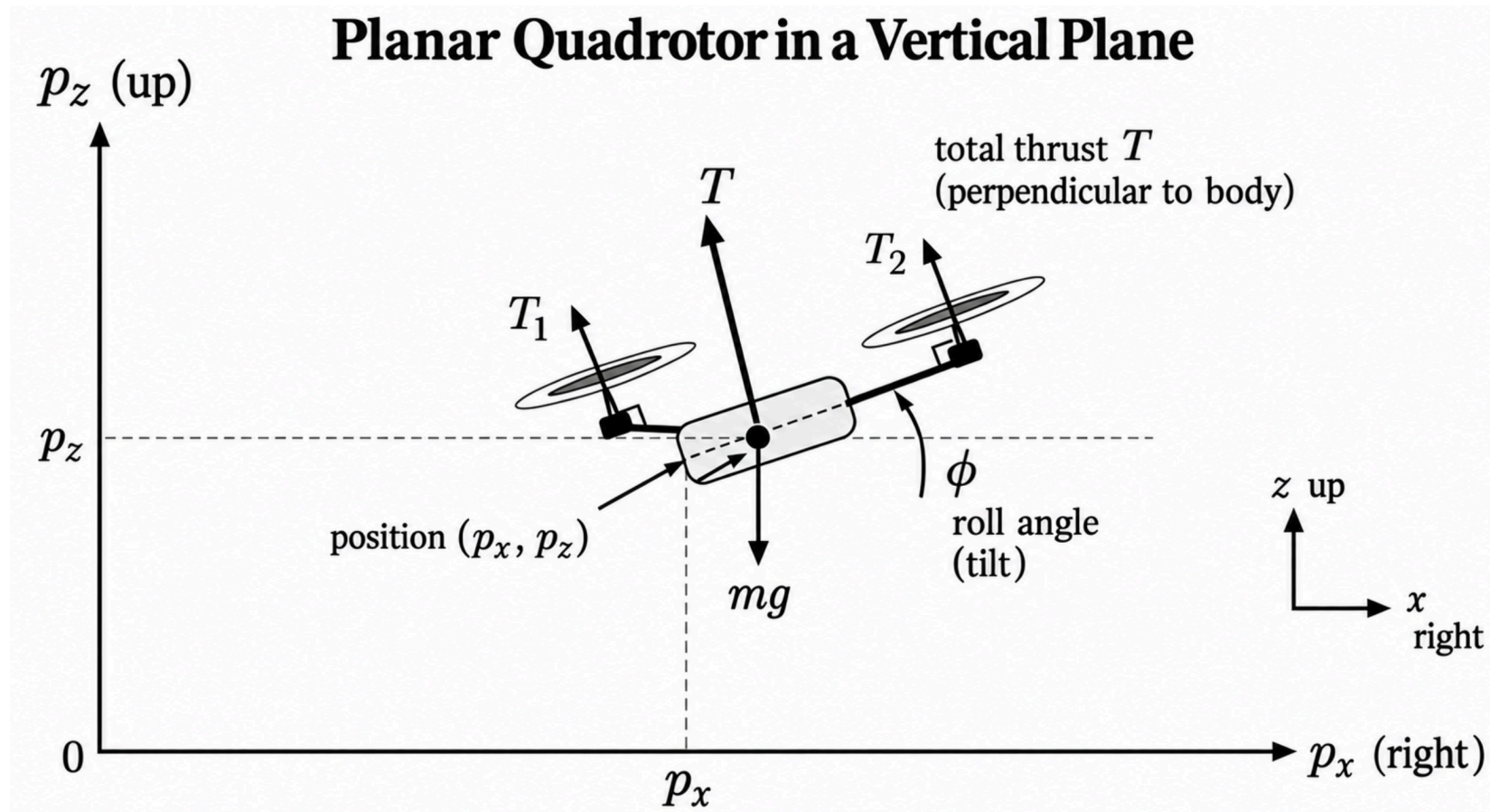


## UNICYCLE: TUPLE

| ELEMENT       | SPECIFICATION                                                                                  |
|---------------|------------------------------------------------------------------------------------------------|
| $\mathcal{X}$ | $(p_x, p_y, \theta, v) \in \mathbb{R}^2 \times S^1 \times \mathbb{R}$ , dim 4                  |
| $\mathcal{O}$ | GPS $(p_x, p_y)$ + wheel encoder $v$ : dim 3 — <b>heading <math>\theta</math> hidden</b>       |
| $\mathcal{A}$ | Forward force + steering torque: $(F, M) \in \mathbb{R}^2$                                     |
| $f$           | $\dot{p}_x = v \cos \theta, \dot{p}_y = v \sin \theta, \dot{\theta} = M/(mR^2), \dot{v} = F/m$ |
| $h$           | $h(p_x, p_y, \theta, v) = (p_x, p_y, v)$ — heading must be inferred from motion history        |
| $\pi$         | To design: path following, obstacle avoidance                                                  |

## PLANAR QUADROTOR: DEFINITION

Quadrotor constrained to a vertical plane.



## PLANAR QUADROTOR: TUPLE

| ELEMENT       | SPECIFICATION                                                                              |
|---------------|--------------------------------------------------------------------------------------------|
| $\mathcal{X}$ | $(p_x, p_z, \phi, \dot{p}_x, \dot{p}_z, \dot{\phi}) \in \mathbb{R}^5 \times S^1$ , dim 6   |
| $\mathcal{O}$ | Barometer $p_z$ + IMU $(\dot{\phi}, \dot{p}_z)$ + optional GPS: dim 3–5                    |
| $\mathcal{A}$ | Total thrust $T$ + moment $M \in \mathbb{R}^2$ (realised by differential rotor speeds)     |
| $f$           | $m\ddot{p}_x = -T \sin \phi, \quad m\ddot{p}_z = T \cos \phi - mg, \quad I\ddot{\phi} = M$ |
| $h$           | Altitude + angular rate commonly observed; horizontal position often partially observed    |
| $\pi$         | To design: hover controller, trajectory tracking                                           |

**In every system:**  $h$  is lossy — velocities are hidden (pendulum, cart-pole), heading is hidden (unicycle), attitude is not directly measured (quadrotor). This is the **generic case**.

# S2 · MECHANICS

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*WHERE DOES THE DYNAMICS COME FROM FOR A MECHANICAL SYSTEM?*



## NEWTON VS LAGRANGE

**Newton's approach:**  $\mathbf{F} = m\mathbf{a}$  — requires summing all forces as vectors.

- Constraint forces (rod tension, joint reactions) must be tracked even though they do no work
- For an  $n$ -joint robot: free-body diagram at every link — bookkeeping grows combinatorially

**Lagrangian approach:** works with **scalars** — kinetic energy  $T$  and potential energy  $V$ .

- Any convenient generalised coordinates (joint angles, arc lengths, quaternions)
- Constraint forces **never appear** — automatically projected out
- For an  $n$ -joint robot: exactly  $n$  scalar equations, one per DOF

## LAGRANGIAN MECHANICS

**Lagrangian:**  $L(q, \dot{q}) = T(q, \dot{q}) - V(q)$

**Definition (Euler–Lagrange equation).** For generalised coordinate  $q$ , velocity  $\dot{q}$ , and generalised force  $Q$ :

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = Q$$

One equation per DOF. Constraint forces drop out automatically.

## DERIVING THE PENDULUM – ENERGIES

**Setup:** mass  $m$ , rod  $l$ , angle  $\theta$  from downward vertical, torque  $\tau$ . So  $q = \theta$ ,  $Q = \tau$ .

Mass position:  $(l \sin \theta, -l \cos \theta)$

**Kinetic energy:**

$$\dot{x}_m = l\dot{\theta} \cos \theta, \quad \dot{y}_m = l\dot{\theta} \sin \theta \quad \Rightarrow \quad T = \frac{1}{2}m|\mathbf{v}|^2 = \frac{1}{2}ml^2\dot{\theta}^2$$

**Potential energy** (pivot as reference, mass height =  $-l \cos \theta$ ):

$$V = -mgl \cos \theta$$

**Lagrangian:**

$$L = T - V = \frac{1}{2}ml^2\dot{\theta}^2 + mgl \cos \theta$$

Sanity check:  $\theta = 0$  (hanging)  $\Rightarrow V = -mgl$  (min)  $\checkmark$      $\theta = \pi$  (inverted)  $\Rightarrow V = +mgl$  (max)  $\checkmark$

## APPLYING EULER-LAGRANGE – EQUATION OF MOTION

$$\frac{\partial L}{\partial \dot{\theta}} = ml^2 \dot{\theta} \Rightarrow \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} = ml^2 \ddot{\theta}, \quad \frac{\partial L}{\partial \theta} = -mgl \sin \theta$$

$$ml^2 \ddot{\theta} - (-mgl \sin \theta) = \tau \quad \Longrightarrow \quad \boxed{ml^2 \ddot{\theta} + mgl \sin \theta = \tau}$$

Dividing by  $ml^2$ , writing  $u = \tau/(ml^2)$ :

$$\ddot{\theta} = -\frac{g}{l} \sin \theta + u$$

**State-space form** with  $x = (\theta, \dot{\theta})^\top$  and  $a = u$ :

$$\dot{x} = f(x, a) = \begin{pmatrix} \dot{\theta} \\ -\frac{g}{l} \sin \theta + a \end{pmatrix}$$

## DISTRIBUTED MASS – MOMENT OF INERTIA (2D)

Real robots have mass distributed throughout their bodies. Different parts are at different distances  $r$  from the rotation axis — so they move at different speeds ( $v = r\dot{\theta}$ ).

**Definition (Moment of inertia, 2D).** For rigid body  $\mathcal{B}$  rotating about a fixed axis:

$$I = \int_{\mathcal{B}} r^2 dm, \quad T_{\text{rot}} = \frac{1}{2} I \dot{\theta}^2$$

$r$  = perpendicular distance from the axis to each mass element  $dm = \rho dV$  ( $\rho$  = volume density, integral is over the body's volume).

For a point mass at distance  $l$ :  $I = ml^2$  — recovering the pendulum case.

## PARALLEL AXIS THEOREM

**Theorem (Parallel axis theorem).** If  $I_{cm}$  is the inertia about an axis through the centre of mass, then for a parallel axis at distance  $d$ :

$$I_{axis} = I_{cm} + md^2$$

**Physical pendulum** (rigid body pivoting at distance  $d$  from centre of mass):

$$I_{pivot} = I_{cm} + md^2 \quad \implies \quad \boxed{I_{pivot} \ddot{\theta} + mgd \sin \theta = \tau}$$

Structure identical to the point-mass case. The equilibria, phase portrait, and stability analysis are unchanged — only the effective inertia differs.

## INERTIA TENSOR (3D) AND KÖNIG'S THEOREM

**Why a scalar  $I$  fails in 3D:** a body resists rotation differently along different axes; angular momentum  $\mathbf{L} = \mathcal{I}\omega$  and angular velocity  $\omega$  are generally **not parallel** — i.e.  $\mathbf{L} \nparallel \omega$ , so spinning a body can produce angular momentum in a direction you didn't push (gyroscopic effect).

**Definition (Inertia tensor, 3D).**

$$\mathcal{I} = \int_{\mathcal{B}} (\|r\|^2 \mathbf{I}_3 - rr^\top) dm, \quad T_{rot} = \frac{1}{2} \omega^\top \mathcal{I} \omega$$

$\mathcal{I}$  is symmetric positive definite. Eigenvectors = **principal axes**: axes about which  $\mathbf{L} \parallel \omega$  (no gyroscopic cross-coupling — spin stays on the axis you chose).

**Theorem (König's theorem).** Kinetic energy splits independently into translational and rotational parts:

$$T = \underbrace{\frac{1}{2} m \|\dot{p}_{cm}\|^2}_{T_{trans}} + \underbrace{\frac{1}{2} \omega^\top \mathcal{I} \omega}_{T_{rot}}$$

The two terms are independent: translational motion does not affect  $T_{rot}$  and vice versa.

## REFERENCE FRAMES

| FRAME          | DEFINITION                                              | PROPERTIES                                                        |
|----------------|---------------------------------------------------------|-------------------------------------------------------------------|
| Inertial frame | Fixed to the world; does not accelerate or rotate       | Newton's laws hold directly: $F = ma$ , $\dot{\mathbf{L}} = \tau$ |
| Body frame     | Fixed to the rigid body; rotates and translates with it | Coordinates are constant for points attached to the body          |

### Why work in the body frame?

The inertia tensor  $\mathcal{I}$  is **constant** in the body frame (the mass distribution doesn't change relative to the body). In the inertial frame,  $\mathcal{I}(t)$  rotates with the body — time-varying and harder to use.

**The price:** time derivatives in the body frame are not the same as in the inertial frame — they pick up a correction due to the frame's rotation  $\omega$ .

## TRANSPORT THEOREM

Newton's  $\dot{\mathbf{L}} = \boldsymbol{\tau}$  holds in an **inertial frame**. The body frame rotates with  $\boldsymbol{\omega}$ , so the time derivative of any vector  $\mathbf{v}$  expressed in body coordinates picks up a correction:

**Theorem (Transport theorem).**

$$\left( \frac{d\mathbf{v}}{dt} \right)_{\text{inertial}} = \left( \frac{d\mathbf{v}}{dt} \right)_{\text{body}} + \boldsymbol{\omega} \times \mathbf{v}$$

**Proof sketch.** Write  $\mathbf{v} = \sum v_i \mathbf{e}_i$  in the body frame. The basis vectors  $\mathbf{e}_i$  are fixed in the body but rotate in the inertial frame:  $\dot{\mathbf{e}}_i = \boldsymbol{\omega} \times \mathbf{e}_i$ . Differentiating:

$$\dot{\mathbf{v}}_{\text{inertial}} = \sum_i \dot{v}_i \mathbf{e}_i + \sum_i v_i (\boldsymbol{\omega} \times \mathbf{e}_i) = \dot{\mathbf{v}}_{\text{body}} + \boldsymbol{\omega} \times \mathbf{v}$$

In 2D  $\boldsymbol{\omega}$  is a scalar, but the correction does **not** vanish. The planar analogue of the cross-product term is  $\omega J \mathbf{v}$ , where  $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  is the  $90^\circ$  counterclockwise rotation matrix.

## EULER'S ROTATIONAL EQUATIONS

Apply the transport theorem to  $\mathbf{L} = \mathcal{I}\omega$  (with  $\mathcal{I}$  constant in the body frame):

$$\tau = \dot{\mathbf{L}}_{\text{inertial}} = \dot{\mathbf{L}}_{\text{body}} + \omega \times \mathbf{L} = \mathcal{I}\dot{\omega} + \omega \times \mathcal{I}\omega$$

**Theorem (Euler's rotational equations).**

$$\mathcal{I}\dot{\omega} + \omega \times \mathcal{I}\omega = \tau_{\text{body}}$$

The term  $\omega \times \mathcal{I}\omega$  is **not** a physical force — it is a frame-change artifact. Physical consequence: **gyroscopic precession** — a torque along one axis produces response perpendicular to both torque and spin.

In 2D:  $\omega \times \mathcal{I}\omega = 0$  (scalar, no cross product)  $\Rightarrow I\ddot{\theta} = \tau$ .

## MASS MATRIX

For an  **$n$ -joint robot arm** with joint angles  $q \in \mathbb{R}^n$ , each link's kinetic energy depends on all joint velocities through the kinematic chain:

$$T = \frac{1}{2} \dot{q}^\top M(q) \dot{q}, \quad M(q) = \sum_{i=1}^n \left[ m_i J_{v,i}(q)^\top J_{v,i}(q) + J_{\omega,i}(q)^\top \mathcal{I}_i J_{\omega,i}(q) \right]$$

where  $J_{v,i}, J_{\omega,i} \in \mathbb{R}^{3 \times n}$  are the translational and rotational Jacobians of link  $i$ 's center of mass, and  $\mathcal{I}_i$  is its body-frame inertia tensor.

$M(q) \in \mathbb{R}^{n \times n}$  is symmetric positive definite, **configuration-dependent**: as the arm changes pose, effective inertia changes.

# MANIPULATOR EQUATION

Applying Euler–Lagrange to  $L = T - V$  gives the **manipulator equation**:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau$$

| TERM                   | NAME                   | ORIGIN                                                                          |
|------------------------|------------------------|---------------------------------------------------------------------------------|
| $M(q)\ddot{q}$         | Inertial               | $\frac{d}{dt} \frac{\partial T}{\partial \dot{q}}$ : resistance to acceleration |
| $C(q, \dot{q})\dot{q}$ | Coriolis / centrifugal | $\dot{M} = \frac{\partial M}{\partial q} \dot{q}$ from product rule             |
| $g(q)$                 | Gravity                | $\frac{\partial V}{\partial q}$                                                 |

**Pendulum check:**  $M = ml^2$  (scalar),  $C = 0$ ,  $g = mgl \sin \theta$

# §3 · PARTIAL OBSERVABILITY

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*WHY A SINGLE OBSERVATION IS ALMOST NEVER ENOUGH*



## WHY OBSERVATION LOSES INFORMATION

- $h$  is almost never injective:  $\exists x \neq x'$  with  $h(x) = h(x')$
- A single  $o_t$  cannot distinguish them — **no algorithm can**

**Jacobian rank test (local injectivity):** compute  $J_h(x_0) = \left. \frac{\partial h}{\partial x} \right|_{x_0} \in \mathbb{R}^{m \times n}$  and check:

**Theorem (Local Injectivity).**  $h$  is locally injective near  $x_0 \iff \text{rank}(J_h(x_0)) = \dim(\mathcal{X})$ .

Conversely, a non-trivial  $\ker J_h$  gives directions invisible to the sensor to first order.

By rank-nullity,  $\dim(\ker J_h) = \dim(\mathcal{X}) - \text{rank}(J_h)$  — this is the **blind subspace** dimension.

If  $\dim(\mathcal{O}) < \dim(\mathcal{X})$ : injectivity **impossible** by rank-nullity, regardless of  $h$ .

This is the common case — sensors almost always observe fewer quantities than the full state. E.g. a rotary encoder on the pendulum gives  $m = 1$  ( $\theta$  only) vs.  $n = 2$  ( $\theta, \dot{\theta}$ ): the velocity is structurally invisible to any single measurement.

# TWO SOURCES OF RANK DEFICIENCY

| CONDITION                                                                        | SOURCE           | CONSEQUENCE                                                           |
|----------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------|
| $\dim(\mathcal{O}) < \dim(\mathcal{X})$                                          | Dimensions alone | Injectivity impossible regardless of $h$                              |
| $\dim(\mathcal{O}) \geq \dim(\mathcal{X}), \text{rank}(J_h) < \dim(\mathcal{X})$ | Structure of $h$ | Injectivity fails near $x_0$ despite sufficient observation dimension |
| $\dim(\mathcal{O}) \geq \dim(\mathcal{X}), \text{rank}(J_h) = \dim(\mathcal{X})$ | —                | Locally injective; single observation can determine $x_t$             |

## EXAMPLE 1 – GPS ON A UNICYCLE

**Dimensional argument** ( $\dim(\mathcal{O}) < \dim(\mathcal{X})$ )

State:  $(p_x, p_y, \theta, v)$ ,  $\dim(\mathcal{X}) = 4$

GPS observation:  $o = (p_x, p_y)$ ,  $\dim(\mathcal{O}) = 2 < 4$

By rank–nullity:  $\dim(\ker J_h) \geq 4 - 2 = 2$

Heading  $\theta$  and speed  $v$  span the **blind subspace** — no matter how the GPS is designed.

No amount of processing of  $o_t$  alone can recover  $\theta$  or  $v$ .

This is structural: it follows from dimensions alone, before inspecting  $h$  at all.

## EXAMPLE 2 – COLLINEAR BEACONS: SETUP

Beacons at  $(0, 0)$ ,  $(1, 0)$ ,  $(2, 0)$ . Range to robot  $(p_x, p_y)$ :  $r_i = \|x - b_i\|$ .

Observation model:  $h(p_x, p_y) = (r_1, r_2, r_3)$ , so  $h : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ .

Each row of  $J_h$  = **unit vector from beacon  $i$  toward the robot** – by differentiating  $r_i = \sqrt{(p_x - b_{ix})^2 + p_y^2}$ :

$$\frac{\partial r_i}{\partial p_x} = \frac{p_x - b_{ix}}{r_i}, \quad \frac{\partial r_i}{\partial p_y} = \frac{p_y}{r_i}$$

$$J_h = \begin{pmatrix} (p_x - 0)/r_1 & p_y/r_1 \\ (p_x - 1)/r_2 & p_y/r_2 \\ (p_x - 2)/r_3 & p_y/r_3 \end{pmatrix}$$

At  $p_y = 0$  (robot **on** the beacon line): all  $y$ -components vanish  $\Rightarrow \text{rank}(J_h) = 1 < 2$ .

## EXAMPLE 2 – COLLINEAR BEACONS: ZERO FIRST-ORDER CHANGE

When  $p_y = 0$ , a perpendicular displacement  $\delta p = (0, \varepsilon)^\top$  is **invisible to first order**. Taylor-expand each range:

$$\begin{aligned} r_i(p_x, \varepsilon) &= \sqrt{(p_x - b_{ix})^2 + \varepsilon^2} = |p_x - b_{ix}| \sqrt{1 + \frac{\varepsilon^2}{(p_x - b_{ix})^2}} \\ &\approx |p_x - b_{ix}| + \frac{\varepsilon^2}{2|p_x - b_{ix}|} + O(\varepsilon^4) \end{aligned}$$

The  $O(\varepsilon)$  term is exactly zero — the first non-zero response is **quadratic** in  $\varepsilon$ :

$$\Delta r_i = r_i(p_x, \varepsilon) - r_i(p_x, 0) = O(\varepsilon^2)$$

With sensor noise  $\sigma$ , the signal  $\Delta r_i \approx \varepsilon^2 / (2|p_x - b_{ix}|)$  is buried in noise until  $\varepsilon \gtrsim \sqrt{2\sigma|p_x - b_{ix}|}$ .

**Physical lesson.** Sensor count alone doesn't guarantee observability — **geometric diversity** matters.  
This is why GPS requires satellites spread across the sky, not clustered in one direction.

## WHAT'S NEXT

### Lecture 3 — Dynamical Systems (II)

- Observability over time · belief state
- Flow & Picard–Lindelöf · equilibria · phase portrait
- Lyapunov stability · linearisation · discretisation (ZOH)